Efficient Taxes on Aviation

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1 Summary

This appendix proposes two new EU own resources on intra-EU flights: a uniform VAT additive to national rates, and an ad valorem tax on vacant seats. Both are collected by the departure state and remitted in full to the EU. Confining the instruments to intra-EU flights is deliberate: it respects the norm against unilateral extraterritorial taxation with revenue retention, avoiding a repeat of the opposition to the 2012 full-scope EU ETS extension and preserving a norm that could underpin stable international financing of global public goods via future tax coalition mechanisms with earmarking obligations.

The vacant-seat tax applies a single statutory rate to the spoilage value of a departure: empty seats counted by cabin, valued at published route- and cabin-specific reference fares drawn from the previous year's realised fares. The flight-level data already exist to an audited standard, leaving fare reporting as the only genuinely new requirement.

The economic case rests on a spoilage distortion. Around 16% of EU seat-kilometres depart empty, closer to 20% weighted by revenue given higher premium-cabin vacancy, and Dörpinghaus et al. (2026) show that market outcomes systematically over-produce vacancy relative to the social optimum, with a vacant-seat tax capable of aligning airline incentives with welfare in the simplest model. There is a social welfare relevant tradeoff between carrying the same passengers with fewer aircraft and raising the risk of early sell-outs that exclude high-valuation travelers, but airlines incentives point towards over-prioritizing keeping seats for high paying late customers.

We present results for an efficiency-maximizing passenger-flow-preserving reform: it sets the vacant-seat tax to reach the efficient vacancy rate, then adjust the additive VAT so passenger numbers are unchanged. Because flows are fixed by construction, the reform places no burden on the tourism sectors. Calibration gives a vacant-seat tax of 82.6% of spoilage value and a 3.7% additive VAT, cutting vacancy from 20.1% to 2.7% and flight frequency by 17.7%. Unusually for a tax reform, it raises EUR 2.5 billion while generating surplus rather than burden: EUR 2.8 billion in partial equilibrium, plus EUR 1.8 billion once resources released from aviation flow to sectors carrying pre-existing taxes and markups, with a further EUR 0.3 billion in oil terms-of-trade gains and EUR 0.5 billion in avoided climate damages accruing to the EU.

Having established that the efficiency-maximizing passenger-flow-preserving reform should be in the interest of all EU countries, we consider a further additional reform that would raise the VAT rate beyond the modest 3.7% implied by that reform. Standard economic theory implies an optimal tax of 28% against an 8% status quo, hence roughly a 20% additive EU VAT.

Finally, the EUR 45 billion base (interval EUR 38–52 billion) is assigned across member states and compared to GNI shares. Malta, Greece, Cyprus, Portugal and Spain are heavily over-exposed, Germany, Slovenia and Slovakia under-exposed. The wide spread implies a standalone VAT increase would likely need compensation, motivating the suggestion to bundle it with the passenger-flow-preserving reform, whose net burden is negative.

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2 The proposals

We propose the following new EU-wide taxes as new EU own resources: 1) A VAT (additive to national VAT rates) on all intra-EU flights; 2) A tax on vacant seats on intra-EU flights.

Both proposed EU own resources are restricted to intra-EU flights, i.e. flights that both depart and land within the EU. This is motivated by the observation that it is valuable to respect the existing international norm against unilateral extraterritorial taxation with revenue retention, both to avoid the kind of opposition faced by the EU during its extension of the EU ETS to full scope in 2012 and to contribute to preserving this norm as it could enable the establishment of stable international mechanisms to fund global public goods via earmarking of aviation tax revenue.

We propose that the entirety of the tax revenue be transmitted to the EU by the country where a flight departs. Under Article 48 of the EU VAT Directive the base of an intra-EU flight is legally fractionated across every Member State overflown. No Member State has a natural claim to the tax base of a border-crossing flight. This motivates the proposed Union-level charge analogous to the EU's customs-duty own resources.

2.1 An additive uniform VAT on intra-EU flights

We propose to implement a VAT on all intra-EU flights with a uniform rate as an EU own resource, with the entire revenue collected by the country of departure and transmitted to the EU. This VAT would be additive to whatever VAT rates that EU member states impose themselves.

2.2 A tax on vacant seats

The instrument is an ad valorem charge on the spoilage value of a flight: the empty seats it departs with, valued at the prevailing market fare for the relevant cabin on the route flown over the entire previous year.

For a flight stage f operated on route r define the spoilage value S_f as follows:

$$S_f = \sum_c (\bar{p}_{r,c} \cdot V_{f,c})$$

where $V_{f,c}$ is the number of vacant seats in cabin c on the stage, $\bar{p}_{r,c}$ is the reference fare — the average fare paid for carriage in cabin c on route r, taken across all carriers operating that route in a past reference period and published in advance.

The spoilage value is the tax basis and a uniform tax rate τ is applied to it. τ is unit-less and thus can be viewed as a kind of ad valorem rate.

The construction separates a quantity, measured at flight level, from a valuation, set at route level. The quantity is observed by the carrier on every departure and is not manipulable. The valuation is a market-wide statistic that the carrier does not influence and knows before it makes the capacity decision.

Vacant seats: the physical base

The base is disaggregated by cabin. Cabin means the physical travel class in which the seat is installed — economy, premium economy, business, first.

Vacant seats in a cabin on a flight stage are the number of passenger seats installed in that cabin of the aircraft as operated on that stage, less the number of persons occupying a seat in that cabin at departure, whether or not those persons paid a fare.

The valuation rule: route-level, cabin-differentiated reference fares

Each vacant seat is valued at the average fare paid for that cabin on that route, across all carriers serving it. Three properties of this rule matter.

A single statutory rate. τ is one number, uniform across routes, cabins and carriers. All heterogeneity — distance, market, seasonality, cabin — enters through the reference fare. The legal instrument fixes one parameter.

The valuation is exogenous to the carrier's own pricing. $\bar{p}_{r,c}$ is a market-wide average from the previous year.

Cabin differentiation. A vacant business seat is valued at the route's average business fare, a vacant economy seat at the route's average economy fare. Beyond proportionality, this is defensible on resource grounds: a long-haul business seat occupies two to three times the floor area of an economy seat, so flying it empty forgoes proportionally more of the aircraft's capacity. It is also necessary for accuracy, since premium cabins typically operate at lower occupancy than economy on the same aircraft; a blended valuation would systematically under-value spoilage where spoilage is highest.

This definition of the tax base is motivated by two considerations, one practical and one welfare-theoretical. On the practical side, the definition has the appeal of converting a physical count into a monetary base that is comparable across routes, cabins, distances and price levels, and that indexes automatically to inflation and fuel costs. On the welfare-theoretic side, results based on Dörpinghaus et al. (2026) explained further below show that the optimal cabin-specific taxation of vacant seats is close to proportional to the cabin-specific spoilage value, i.e. as under the proposed tax base.

Constructing and governing the reference-fare schedule

Coverage. The reference fare for a cell is computed over all carriers operating the route in the reference period, weighted by passengers. It is a market statistic, not a carrier statistic; a given carrier's own fares influence it only through its market share.

Timing. The schedule is fixed for a calendar year, computed from realised fares over the twelve months prior to departure.

Fare basis. The reference fare is the total amount paid by the passenger for carriage on the stage, net of taxes and of government and airport charges collected on behalf of third parties, but inclusive of ancillary charges for seat selection, baggage and fare bundles.

Fare reporting. The European Union has no public route-level fare database comparable to the United States DB1B origin-and-destination survey. Constructing the schedule requires either a new reporting obligation on carriers covering passengers and passenger revenue by route, cabin and period — figures carriers already hold — or the procurement of commercial fare data.

Cabin definitions. "Business class" is not standardised across carriers or markets. The schedule requires a harmonised classification, defined by reference to the operator's declared cabin designation in its schedule filing, with an anti-abuse provision preventing reclassification undertaken principally to alter the applicable reference fare. If such a provision is judged to be difficult to make effective, then the French solidarity levy on air travel's binary classification of cabins could be used instead, where *a seat falls in the additional-services category where a passenger carried in that seat would be entitled, without supplement, to on-board services that not all passengers on the flight can access without one.* This is determinate from the cabin configuration and the carrier's published product rules and requires no new taxonomy.

Measurement, reporting and verification

The quantity is already generated, to an audited standard, on every commercial flight. Mass-and-balance documentation prepared under EU air operations rules (Regulation (EU) No 965/2012) records the number of persons on board at departure; this figure is safety-critical, produced by the departure control system at door close, and retained by the operator. Installed seats by cabin are fixed by the certified cabin configuration recorded for the airframe.

The reporting obligation is therefore modest: for each in-scope stage, the operating carrier reports (i) installed passenger seats by cabin in the configuration flown, and (ii) persons occupying a seat by cabin at departure. Spoilage value and liability follow mechanically by applying the published schedule. Returns are filed monthly and verified against records the carrier is already required to hold. No new measurement is

created at flight level; the only genuinely new data requirement is the revenue reporting needed to construct the schedule itself.

3 The economic case for the tax on vacant seats

Currently, 16% of seat-kilometres on flights remain empty in the EU. Weighted by revenue shares, this number is close to 20% given that business class seats have a substantially higher vacancy rate. This raises the question of whether this is an efficient allocation of resources.

To analyze this question, it is useful to distinguish two logical possibilities. Either the cause for the vacant seats is "deterministic" or "stochastic". By "deterministic" we shall here mean that at some point in time (early in its sales horizon) before departure of a flight it becomes certain that the flight will not sell out. In a way, it is implausible that there is widespread such deterministic vacancy. After, all, why wouldn't airlines rent smaller airplanes? However, in periods when there are structurally asymmetric passenger flows between two places, deterministic vacancy can happen.

If one could design tax policy applying specifically to cases of such deterministic vacancy, the optimal policy prescription would be straightforward: if there is no scarcity then seats should only be priced at marginal cost, i.e. to close to 0. Hence if prices are above marginal costs then increasing a tax on vacant seats will cause increased utilization valued higher than the marginal cost, thereby raising social welfare.

Now consider the case where the cause for the vacant seats is stochastic, i.e. where seats remain empty only if by chance demand turns out to be low. In this case, there is a genuine tradeoff involved in taxing vacant seats. On the one hand, it will cause additional people to take the flight. On the other hand, it will increase the probability of early sell-outs and therefore of people with very high valuations for taking the flight not being able to take it.

For the purpose of analyzing the passenger-flow-preserving tax reforms (see next section), we can phrase the tradeoff as follows: taxing vacant flights and appropriately adjusting the VAT allows for transporting the same number of people with fewer planes which saves resources. However, it also makes it more likely that flights sell out before departure so that people with very high willingness to pay will not get a ticket.

Dörpinghaus et al. (2026) shows that there is a systematic "spoilage distortion": without corrective tax policy, market outcomes will involve an excessively large vacancy rate (from a social welfare perspective). Moreover, Dörpinghaus et al. (2026) show that in the simplest model and a broad range of forms of travel demand, a tax on vacant seats can perfectly align a private airline's incentive with the objective of social welfare. These results provide initial motivation for the tax on vacant seats. In the next section we will provide quantitative results about the welfare gains that could be reaped, using a full-fledged model introduced in the second part of Dörpinghaus et al. (2026) .

4 Passenger-flow-preserving tax reforms

We propose the following tax reform as a reform that we will argue is robustly beneficial:

Definition 1 *The efficiency-maximizing passenger-flow-preserving tax reform consists of introducing the two proposed tax instruments, i.e.:*

1) An additive[1] *intra-EU uniform VAT.*

2) A tax on vacant seats, as defined above, i.e. with a uniform ad valorem rate applied to the spoilage value of empty seats.

The reform consists of setting the tax on vacant seats to achieve that the socially efficient vacancy rate gets reached and adjusting the VAT tax such that the number of passengers being transported remains unchanged as compared under the status quo.

4.1 Evaluating passenger-flow-preserving tax reforms

We now provide results about the efficiency-maximizing passenger-flow preserving tax reforms that we computed using the partial equilibrium model developed in Dörpinghaus et al. (2026), combined with a welfare

[1] i.e. additive to other taxes

accounting adjustment derived in Stern (2026) to capture the fact that there are pre-existing taxes and markups on the other goods in the economy. The calibrated model yields the following results for the efficiency-maximizing passenger-flow-preserving reform which we will explain below:[2]

Table 1.1: Policy Parameters and Summary of Impacts for the efficiency-maximizing passenger-flow-preserving tax reform applied to all intra-EU flights

Policy Parameters	
Tax on vacant seats	82.6% of spoilage value
Additive EU-wide VAT rate[a]	3.7%
Flight frequency reduction	17.7%
Impacts and Outcomes[b]	
Vacancy rate (from 20.1%)	2.7%
Emissions reduction (excl. leakage)[c]	$\approx$8.0 $MtCO_2$
Emissions reduction (incl. leakage)[c]	$\approx$1.76 $MtCO_2$
Aviation tax revenue increase	+EUR 2.5 billion
Economic surplus change (excl. climate damages)	+EUR 2.8 billion
incl. cross-sector wedge correction[d]	+EUR 4.6 billion
Global Climate Damages Avoided[c]	
Avoided climate damages from CO_2 (SCC EUR 180/t)	EUR 1.5 billion / year
Avoided climate damages incl. non-CO_2 forcing, i.e. contrails (SCC EUR 290/t)	EUR 2.3 billion / year
Terms of trade effects from reduced oil demand[c]	
Savings for the EU's oil import bill	EUR 300 million / year
Reduction in oil revenue for Russia	EUR 203 million / year
Cross-Sector Fiscal Spillovers[d]	
Additional consumption-tax revenue (11.5%)	EUR 0.75 billion
Additional corporate tax on markups (26% $\times$ 14.5%)	EUR 0.25 billion

Notes: Results are for the passenger-flow-neutral policy package. [d] Applies the pre-existing wedge of $1.115 \times 1.145 - 1 = 0.28$ in the rest of the economy to the resource savings released by the 17.7% reduction in flight frequency (EUR 6.5 billion).

The findings are striking: the tax reform collects EUR 2.5 billion in additional revenue from the aviation sector. Rather than creating an economic burden in the vicinity of the tax revenue collected, as is typical of tax reforms, it actually generates an economic surplus of EUR 4.6 billion. This results is obtained by in a first step computing the economic surplus from the reform in the partial equilibrium model from Dörpinghaus et al. (2026) which comes out as EUR 2.8 billion. The result qualitatively confirms the intuitive appeal of the proposed tax reform: it saves resources while not reducing the amount of passenger flows. The partial equilibrium analysis misses a further effect that a simple result from Stern (2026) allows us to account for in a fully rigorous way as a second step: the resources saved in the aviation sector get redirected to other sectors. Given that other sectors have pre-existing commodity taxes and markups, this leads to additional profits and tax revenue. Overall, this brings in an additional EUR 1.8 billion in welfare gains of which EUR 0.75 billion takes the form of additional consumption tax revenue and EUR 0.25 billion taking the form of additional corporate income tax revenue. In addition, the EU reaps EUR 0.3 billion in terms of trade benefits (based on Beaufils et al. (2025)) and, assuming its SCC share equals its GDP share, EUR 0.5 billion in avoided climate damages. Hence overall, the EU reaps EUR 5.4 billion in economic surplus from the EUR 2.5 billion in additional revenue.

Incidence

In general, aviation tax reforms raise the difficult question of how the incidence is split between the travelers and the hospitality sector. Conceptually, it is clear that this is determined by the relative elasticities of the demand for flights and the supply of hospitality services (Black et al. (2024)). However, estimating these elasticities is difficult and, as a result, there appear to be no published studies giving any confident estimates.

[2]The model disregards distributional questions to focus on efficiency. However, taking into account distributional effects would typically strengthen the case for taxing vacant seats.

Luckily, to discuss the incidence of the passenger-flow-preserving tax reforms, we do not need to take a stance on these questions.

In fact, an attractive feature of the passenger-flow-preserving tax reforms is that by design, they create no burden for the hospitality sector, i.e. for those providing goods and services to travelers such as hotels, restaurants and tourism service providers. Moreover, the impact on travelers (via changes in prices, changes in availability of flights and changes in flight frequency) is well-captured by the model from Dörpinghaus et al. (2026) and thus already taken into account in the above estimates.

Taken together, our results imply that all EU countries should find it in their interest to vote in favor of the efficiency-maximizing passenger-flow-preserving tax reform.

5 The economic case for the additive uniform VAT on intra-EU flights

So far we have focused on passenger-flow-preserving tax reforms. However, now we shall consider an additional tax reform, consisting of raising the EU-wide additive uniform VAT beyond the modest value corresponding to the efficiency-maximizing passenger-flow-preserving tax reform.

The basic case for such an additional tax reform is that aviation is undertaxed relative to other consumption goods. In fact, a broad class of economic models imply that optimal taxation requires equalization of "wedges" across all consumption goods (Atkinson and Stiglitz (1976), Epifani and Gancia (2011)). Here the "wedge" denotes the proportion by which the consumer price exceeds the marginal cost of the good. Hence, by definition, we have $1 + wedge = (1 + tax)(1 + markup)$, where "markup" denotes the proportion by which the firms set the price above marginal cost.

This result yields a simple heuristic for identifying efficiency increasing tax reforms when a given sector (e.g. aviation) is under consideration in isolation: setting the tax rate such that the wedge equals the (appropriately weighted) average of the wedges in the rest of the economy. In the EU, the average markup on non-aviation consumer goods and services is 11.5 percent and the average tax rate is 14.5 percent. Air travel is a special sector in the sense that markups can be taken to be 0, explained by the ease of entry for airlines on any route. These numbers imply that a tax on flights of $(1 + 0.115)(1 + 0.145) - 1 = 0.28\%$ of the ticket price would be optimal. Given the 8% status quo average intra-EU ad valorem tax rate from ticket taxes and domestic VATs, this suggests that an additive EU-wide VAT of 20% would be optimal from the EU's perspective (not taking into account that externalities and terms-of-trade effects would justify even higher rates).

Optimal taxation theory has also found that all intermediate inputs to production should be tax exempt (Diamond and Mirrlees (1971)), exactly as ensured by any VAT. This provides an efficiency rationale for replacing existing ticket taxes with a VAT.

Thus standard economic theory suggests that in addition to the 4% EU-wide additive VAT that is part of the efficiency-maximizing passenger-flow-preserving reform, there are two further reforms that would increase efficiency: 1) a replacement of ticket taxes by an ad-valorem equivalent VAT increase, 2) an additional increase in the VAT by 16%.

6 Incidence of the VAT

For the passenger-flow preserving tax reforms that are the core proposal of this appendix, we saw that the incidence analysis got greatly simplified. However for the further increase in the additive EU-wide VAT on intra-EU flights, one would need to calibrate a model with endogenous tourism supply to be able to compute how the tax incidence is distributed over tourism service suppliers and travelers. Such a rigorous economic incidence analysis at the country level is beyond the scope of this report. However, in the next subsection, we provide an estimate for the proportions of the tax base corresponding to the different member states. This gives us a rough proxy for the proportions of the incidence born by these states. The previous section's analysis gives us an indication of the cost of raising public funds via the VAT: at an EU-wide additive VAT rate of 20%, the marginal cost of public funds raised via it is 1 and for lower rates of the EU-wide additive

VAT it will be lower[3].

6.1 VAT tax base estimates by EU member states

Suppose we take the assigned tax base as a proxy for the economic burden of the tax and suppose that EU own resource will displace GNI-based contributions one for one. Then the ratios of the share of tax revenue to the share of GNI constitute an indicator on whether a country is disproportionately burdened.

After the efficiency-maximizing passenger-flow-preserving reform, we found there to be an under-taxation by 20%. Under the perspective of Jacobs (2018), this implies a marginal cost of public funds of 0.9 if one assumes a price elasticity of demand of -0.6 and an infinite price elasticity of supply. If the average cost of raising a EUR via the VAT on flights is 0.9 then a country is better off without this new own resource as long as its ratio of the share of the tax base to the share of the GNI exceeds $1/0.9 = 1.11$. The fact that several countries have a ratio greatly exceeding 1 suggests that reaching unanimity might require some compensation scheme if a VAT increase is put up as a separate proposal for an EU own resource. Alternatively, the efficiency-maximizing passenger-flow-preserving tax reform described above could be bundled with a further increase in the VAT and be proposed as a package of EU new own resources. Given our result that the former reform would actually have negative net tax burden, this path should be explored. The results provided in this report provide a basis on which one could analyze how far this approach could go.

[3]This proxy likely overestimates the true burden substantially, given that a part of the incidence actually falls on non-EU travelers and the non-EU hospitality sector. Again, due to data limitations, quantifying this effect is beyond the scope of this report.

Table 1.2: Assignment of the intra-EU aviation VAT tax base to member states, 2024

					Revenue relative to GNI (bp)	
Member state	Tax base (EUR bn)	Share of base (%)	Share of EU GNI (%)	Ratio	Passenger-flow-preserving Reform (EUR 2.5bn)	Additional VAT of 20% (EUR 8bn)
Malta	0.27	0.61	0.11	5.60	7.59	24.29
Greece	2.54	5.64	1.30	4.34	5.88	18.82
Cyprus	0.29	0.64	0.16	3.90	5.29	16.92
Portugal	1.95	4.33	1.55	2.80	3.80	12.14
Spain	9.33	20.74	8.68	2.39	3.24	10.37
Croatia	0.38	0.84	0.48	1.76	2.39	7.63
Latvia	0.18	0.40	0.24	1.67	2.26	7.24
Luxembourg	0.21	0.46	0.31	1.48	2.01	6.42
Italy	7.11	15.81	12.06	1.31	1.78	5.68
Denmark	1.13	2.50	2.27	1.10	1.49	4.77
Bulgaria	0.27	0.60	0.55	1.09	1.48	4.73
Finland	0.72	1.59	1.60	1.00	1.36	4.34
Estonia	0.09	0.22	0.22	0.96	1.30	4.16
Ireland	0.94	2.10	2.33	0.90	1.22	3.90
Austria	1.09	2.43	2.81	0.86	1.17	3.73
Sweden	1.16	2.58	3.23	0.80	1.08	3.47
Lithuania	0.14	0.31	0.42	0.74	1.00	3.21
Netherlands	1.98	4.40	6.06	0.73	0.99	3.17
France	5.38	11.96	16.61	0.72	0.98	3.12
Belgium	1.10	2.45	3.46	0.71	0.96	3.08
Hungary	0.37	0.82	1.18	0.70	0.95	3.04
Romania	0.62	1.37	2.03	0.68	0.92	2.95
Czechia	0.44	0.98	1.68	0.58	0.79	2.52
Poland	1.19	2.65	4.77	0.56	0.76	2.43
Germany	6.04	13.43	24.81	0.54	0.73	2.34
Slovenia	0.03	0.06	0.38	0.16	0.22	0.69
Slovakia	0.05	0.10	0.73	0.14	0.19	0.61
EU27	45.00	100.00	100.00	1.00	1.36	4.34

Notes: The tax base is pre-tax consumer spending (fares plus ancillary revenue, net of existing ticket taxes and VAT, gross of airport, security and air navigation charges) on scheduled passenger flights with both endpoints in the EU27, including domestic flights, less the share attributable to business travel on the grounds that VAT-registered businesses reclaim input VAT. The aggregate base is EUR 45bn, with an estimation interval of EUR 38–52bn. Domestic flights are assigned in full to the member state concerned; cross-border intra-EU flights are assigned half to the departure country and half to the arrival country. Passenger volumes are from Eurostat (`avia_paoc`, 2024), for which the national and intra-EU aggregates are constructed from departure declarations only and are therefore a census of one-way passenger legs; revenue per passenger and the business-travel share are estimated and carry essentially all of the uncertainty. GNI shares are the own-resources key agreed by the Advisory Committee on Own Resources and reproduced in the Statement of Revenue of the Draft Budget 2025. The ratio is the share of the assigned tax base divided by the share of EU GNI, computed from unrounded shares. Member states are ordered by that ratio.

The final two columns give revenue collected in each member state as a proportion of that member state's GNI, in basis points (1bp = 0.01% of GNI); divide by 100 to read as a percentage. Column 6 is the revenue collected from the passenger-flow-preserving reform, EUR 2.5bn distributed in proportion to the assigned tax base; column 7 is the revenue collected additionally from a further additive VAT increase of 20 percentage points, EUR 8bn distributed on the same key. GNI levels are taken from Table 4 of the Statement of Revenue of the Draft Budget 2025, for which the EU27 GNI base is EUR 18,445bn.

7 The case for restricting the scope of aviation taxes to intra-EU flights

7.1 Avoiding international opposition and trade sanctions

The EU's experience in 2012 provides strong evidence on the political risks of unilateral extraterritorial aviation taxation with full revenue retention: Directive 2008/101/EC brought all flights arriving in or departing from EU airports into the ETS as of 1 January 2012. On 22 February 2012, twenty-three states adopted the Moscow Declaration, whose annex set out candidate countermeasures: proceedings under Article 84 of the Chicago Convention, prohibition of their own carriers from participating in the scheme, an assessment of the ETS against the EU's WTO commitments, review of air services agreements, suspension of negotiations on new routes, and additional levies on EU carriers. China's civil aviation authority had already, on 6 February 2012, forbidden Chinese carriers from participating without prior approval, and India followed. Airbus reported in March 2012 that thirty-five A330 and ten A380 orders — roughly $12 billion at list prices — were being withheld by China, and that it faced retaliation threats originating in twenty-six countries (EADS statements, March 2012). The Commission announced the "stop the clock" suspension on 12 November 2012, one day before the US Congress passed the EU ETS Prohibition Act, subsequently signed into law that month.[4] The measure had thus survived only eleven months.

At the core of this diplomatic difficulty is a legitimacy problem inherent in unilateral extra-territorial taxation with full revenue retention. Economically, the norm against unilateral extra-territorial taxation with revenue retention is well-founded in the sense that much of the burden of taxes on international flights is born by citizens from other countries and without . For the purposes of defining an EU-own resource, it is therefore best to restrict the scope to intra-EU flights. Taxes on flights reaching beyond the EU should only be introduced in combination with earmarking of the revenue for global funds.

7.2 Preserving the option of establishing an aviation levy coalition mechanism with earmarking for global funds for global public goods

The EU could initiate an aviation levy coalition mechanism where coalition members would tax all flights landing or departing on their territory but earmark this revenue to global funds legitimized as being in the global general interest via some voting procedure (e.g. at the UN). Coalition members could decide how to allocate the tax revenue that they collect to the different global funds. If all countries participate in such a mechanism then none would have a strong incentive to leave, given that its flights connecting it to the other countries would be taxed anyway and given that by being part of the coalition a country can influence through its earmarking the allocation of money across the different global funds.

The current norm against unilateral extra-territorial taxation of flights with revenue retention creates a favorable situation for an aviation levy coalition: the norm has achieved that flights are taxed well below what would arise from unrestrained unilateral taxation. Thus participation in an aviation levy coalition mechanism is attractive. The mechanism has the potential to generate a stable source of funding for global funds.

It follows that it is important to preserve the global norm against unilateral extra-territorial taxation with revenue retention. For this it is essential for the EU to respect the norm. If it were to act in violation of it, as it did for 11 months in 2012, and not yield to international pressure then the norm could get eroded and other countries might follow suit in also introducing unilateral extra-territorial taxes on flights and retain the revenue. The opportunity to instead establish an aviation levy coalition mechanism would have been lost.

[4]The suspension was enacted as Decision No 377/2013/EU and has been extended by successive instruments; the reduced scope covering only intra-EEA flights remains in force today.

Bibliography

Atkinson, A. B. and Stiglitz, J. E. (1976). The design of tax structure: Direct versus indirect taxation. *Journal of Public Economics*, 6(1–2):55–75.

Beaufils, T., Conyngham, K., de Vries, M., Jakob, M., Kalkuhl, M., Richter, P. M., Spiro, D., Stern, L., and Wanner, J. (2025). The geopolitical externality of climate policy. Kiel Working Paper 2283, Kiel Institute for the World Economy.

Black, S., Parry, I. W. H., Singh, S., and Vernon-Lin, N. (2024). Destination net zero: The urgent need for a global carbon tax on aviation and shipping. IMF Staff Climate Note 2024/003, International Monetary Fund.

Diamond, P. A. and Mirrlees, J. A. (1971). Optimal taxation and public production i: Production efficiency. *American Economic Review*, 61(1):8–27.

Dörpinghaus, A., Kalkuhl, M., and Stern, L. (2026). Efficient aviation tax policy with endogenous vacancy rates. available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5076894.

Epifani, P. and Gancia, G. (2011). Trade, markup heterogeneity and misallocations. *Journal of International Economics*, 83(1):1–13.

Jacobs, B. (2018). The marginal cost of public funds is one at the optimal tax system. *International Tax and Public Finance*, 25(4):883–912.

Stern, L. (2026). Wedges and welfare: Exact surplus accounting for sector-specific reforms in a quasilinear economy with a distorted numeraire. Unpublished manuscript, available at https://drive.google.com/file/d/1Duv23AccjhuEd2300bUrlNpuBa28hwjG/view?usp=sharing.